

TESTING OF HYPOTHESIS:

Note:

1. Sample size - n
2. Population size - N
3. Sample mean - $\bar{x}$
4. Sample variance - s^2
5. Population mean - μ
6. Population variance - σ^2

NULL HYPOTHESIS:

* For applying the test of significance the first step is a hypothesis which is definite statement about the population parameter, usually such a hypothesis is a no difference and it is denoted by H_0

ALTERNATIVE HYPOTHESIS:

* Any hypothesis which is complementary to the null hypothesis is called an alternative hypothesis.

* Denoted by H_1 .

TYPE I AND TYPE II ERROR:

type I error: when H_0 is true, we may reject it.

type II error: when H_0 is wrong, we may accept it.

That is accept H_0 when H_1 is true.

Critical values (Z_α) of Z :

	Level of significance.		
	1% (0.01)	5% (0.05)	10% (0.1)
Two tailed test $\mu \neq \mu_0$	$ Z_\alpha = 2.58$	$ Z = 1.96$	$ Z = 1.645$
Right tailed test $\mu < \mu_0$	$Z_\alpha = 2.33$	$Z_\alpha = 1.645$	$Z_\alpha = 1.28$
Left tailed test $\mu > \mu_0$	$Z_\alpha = -2.33$	$Z_\alpha = -1.645$	$Z_\alpha = -1.28$

Test of significance for large samples.

1. Testing of significance for single mean.

2. Testing of significance for difference of mean.

3. Testing of significance for single proportion

4. Testing of significance for single proportion

14. Testing of significance for difference of proportion.

①. Testing of significance for single mean.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$V(\bar{x}) = \frac{s^2}{n}$$

$$E[s^2] = \frac{n-1}{n} \sigma^2$$

$$E(s^2) = \sigma^2, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$Z = \frac{\bar{x} - \mu}{(\sigma/\sqrt{n})}$$

$$Z = \frac{\bar{x} - \mu}{(s/\sqrt{n})}$$

15. The average marks in mathematic of a sample of 100 students was 51 with standard deviation of 6 marks, could this have been a random sample for a population with average mark 50?

Sol:

$H_0: \mu = 50$. If we can't find value for H_0

we can write no significant

Given,

$H_1: \mu \neq 50$

differs.

Sample size $n = 100$

Sample mean $\bar{x} = 51$,
Sample S.D $s = 6$.

$$Z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{51 - 50}{6/\sqrt{100}}$$

$$= \frac{1}{6/10}$$

$$= \frac{10}{6}$$

$$Z = \frac{5}{3} = 1.666$$

we choose 1% significant difference.

[two tailed test]

$$|Z_{1\%}| = 2.58 \Rightarrow \text{table value}$$

$$|Z| = \text{calculated value} = 1.666$$

$$|Z| < |Z_{1\%}|$$

Hence, accept H_0 .

16. The heights of college student in a city are normally distributed with S.D 6 cm. A sample of 100 students has mean height 158 cm. Test the hypothesis that the mean height of college student in a city is 160 cm.

Sol:

Given,

$\sigma = 6$
 $n = 100$

$H_1: \mu \neq 160$

$H_0: \mu = 160$

$$\bar{x} = 158$$

$$n = 100$$

$$Z = \frac{\bar{x} - \mu}{(\sigma/\sqrt{n})} = \frac{158 - 160}{(6/\sqrt{100})}$$

$$= \frac{-2}{6/10}$$

$$Z = \frac{-10}{3} = -3.333$$

$$|Z| = 3.333$$

we choose 1% and 5% level of significance.

$$|Z_{1\%}| = 2.58, |Z_{5\%}| = 1.96$$

$$|Z_{1\%}| < |Z|$$

$$|Z_{5\%}| < |Z|$$

Hence reject H_0 .

type-II error (μ , true)

3. A test was given to a large group of boys who scored an average 64.5 marks. Same test was given to a group of 100 boys whose score an average of 65.5 marks, with a S.D. 10.5 marks. Examine if the difference is significant.

Sol.

$H_0: \mu =$ no significant difference.

$\mu \neq$ significant difference.

Given,

$$\mu = 64.5$$

$$n = 400$$

$$\bar{x} = 65.5$$

$$S = 10.5$$

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{65.5 - 64.5}{10.5/\sqrt{400}}$$

$$= \frac{2 \times 20}{10.5}$$

$$= \frac{40}{10.5}$$

$$|Z| = 3.8$$

we choose 1% & 5%

$$|Z_{1\%}| = 2.58, |Z_{5\%}| = 1.96$$

$$|Z_{1\%}| < |Z|$$

$$|Z_{5\%}| < |Z|$$

Hence reject H_0 .

4. The mean breaking strength of the cables supplied by a manufacturer is 1800 with a standard deviation of 100 by a new technique in the manufacturing

process. It is claimed that the breaking strength of cable has increased in order to test this claim a sample of 50 cables is tested and it is found that the mean breaking strength is 1850 can we support the claim at 2% level of significance.

$$H_0: \mu = 1800$$

$$H_1: \mu \neq 1800$$

Sol: Given, $\mu = 1800$

$$\sigma = 100$$

$$n = 50$$

$$\bar{x} = 1850$$

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{1850 - 1800}{100/\sqrt{50}}$$

$$Z = \frac{50 \times 7.0710}{100}$$

$$Z = \frac{7.0710}{2}$$

$$|Z| = 3.5355$$

$$|Z_{1-\alpha/2}| = 2.58$$

$$|Z_{1-\alpha/2}| < |Z|$$

Hence reject H_0 .

③. Testing of significance for difference of mean of two large sample.

The test statistics is

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Note:

Under $H_0: \mu_1 = \mu_2$ if the samples are drawn from the same population.

where, $\sigma_1 = \sigma_2 = \sigma$

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Note:

If σ_1, σ_2 are not known in equation and $\sigma_1 \neq \sigma_2$ the test statistic used in this case is

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

57. In a Random sample of size 500, the mean is found to be 20. In another independent sample of size 400, the mean is 15. Could the samples have been drawn from the same population with S.D. = 4.

Given,

$$n_1 = 500, n_2 = 400$$

$$\bar{x}_1 = 20, \bar{x}_2 = 15$$

$$\sigma_1 = \sigma_2 \Rightarrow \sigma = 4$$

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$H_0 : \sigma_1 = \sigma_2$$

$$H_1 : \sigma_1 \neq \sigma_2$$

$$Z = \frac{20 - 15}{4 \sqrt{\frac{1}{500} + \frac{1}{400}}}$$

$$= \frac{5}{4 \times \sqrt{4.5 \times 10^{-3}}}$$

$$= \frac{5}{4 \times 0.0670}$$

$$= \frac{5}{0.2683}$$

$$\boxed{|Z| = 18.633}$$

we choose $|Z_{1\%}|$ & $|Z_{5\%}|$ level of significant.

$$|Z_{1\%}| = 2.58$$

$$|Z_{5\%}| = 1.96$$

$$|Z_{5\%}| < |Z|$$

Hence reject H_0 .

The samples are not drawn from same population ($\sigma = 4$).

A sample of heights. 6400 english men as a mean of 170 cm and a S.D of 6.4 cm while a sample of heights of 1600 Americans as a mean of 172 cm and a S.D of 6.3 cm, do the data indicate the americans are on the average taller than the english men?

Sol:

$$n_1 = 6400, n_2 = 1600$$

$$\bar{x}_1 = 170 \text{ cm}, \bar{x}_2 = 172 \text{ cm}$$

$$s_1 = 6.4 \text{ cm}, s_2 = 6.3 \text{ cm}$$

$$H_0 : \bar{x}_1 < \bar{x}_2 \quad (\text{Right tail})$$

$$H_1 : \bar{x}_1 > \bar{x}_2$$

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{170 - 172}{\sqrt{\frac{(6.4)^2}{6400} + \frac{(6.3)^2}{1600}}}$$

$$= \frac{-2}{\sqrt{\frac{40.96}{6400} + \frac{39.69}{1600}}}$$

$$= \frac{-2}{\sqrt{0.03120}} = \frac{-2}{0.1766}$$

$$Z = -11.3216$$

$$|Z| = 11.3216$$

Let us choose 1% level significant.

$$|Z_{1\%}| = 2.33$$

$$|Z_{1\%}| < |Z|$$

$\therefore$ reject H_0 .

$\therefore$ The English men are on the average taller than the Americans.

7) Test the significance of the difference b/w the means of the samples drawn from two normal populations with the same S.D from the following data.

	Size	Mean	S.D
Sample 1	100	61	4
Sample 2	200	63	6

Sol:

$$n_1 = 100 \quad n_2 = 200$$

$$\bar{x}_1 = 61 \quad \bar{x}_2 = 63$$

$$s_1 = 4 \quad s_2 = 6$$

$$H_0: s_1 = s_2$$

$$H_1: s_1 \neq s_2$$

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$Z = \frac{61 - 63}{\sqrt{\frac{16}{100} + \frac{36}{200}}}$$

$$Z = \frac{-2}{0.5830}$$

$$Z = -3.4299$$

$$|Z| = 3.4299$$

Let us choose 1% & 5% level of significant.

$$|Z_{1\%}| = 2.58 \quad |Z_{5\%}| = 1.96$$

$$|Z_{1\%}| < |Z| \quad |Z_{5\%}| < |Z|$$

$\therefore$ Hence rejects H_0 .

8) The average marks score by 35 boys is 72 with a S.D of 8 and 36 girls is 70 with S.D of 6. Test at 1% level of significance whether the boys perform better than girls.

Sol:

$$\text{Given, } n_1 = 35$$

$$n_2 = 36$$

$$\bar{x}_1 = 72$$

$$\bar{x}_2 = 70$$

$$s_1 = 8$$

$$s_2 = 6$$

$$H_0: \bar{x}_1 > \bar{x}_2 \quad (\text{left tail})$$

$$H_1: \bar{x}_1 < \bar{x}_2$$

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$= \frac{72 - 70}{\sqrt{\frac{64}{32} + \frac{36}{36}}}$$

$$= \frac{2}{\sqrt{2+1}}$$

$$= \frac{2}{\sqrt{3}} = \frac{2}{1.732}$$

$$|Z| = 1.154$$

$$|Z_{1\%}| = 2.58$$

$$|Z_{1\%}| < |Z|$$

∴ Hence accepts H_0 .

⑧ Testing of significance for single proportion

p = observed proportion, P = population proportion

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} \quad [P+Q=1]$$

Note:

Confident limits for observed population proportion

$$p \text{ are } p \pm 3 \sqrt{\frac{PQ}{n}}$$

Confident limits for population proportion

$$P \text{ are } P \pm 3 \sqrt{\frac{PQ}{n}} \quad p \pm 3 \sqrt{\frac{PQ}{n}}$$

a) A coin is tossed 556 times and 132 heads are obtained. Could you conclude that the coin is a biased one?

H_0 = A coin is unbiased.

H_1 = A coin is biased.

Sol:

Given,

$$p = \frac{132}{556} = 0.2356$$

$$P = \frac{1}{2} = 0.5$$

$$Q = 1 - P = 1 - 0.5 = 0.5$$

$$n = 556$$

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$$

$$= \frac{0.2356 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{556}}}$$

$$= \frac{0.2644}{\sqrt{9.7656 \times 10^{-4}}}$$

$$= \frac{0.2644}{0.03125}$$

$$|Z| = 8.464$$

Let us choose 1% & 5%.

$$|Z_{1\%}| = 2.58 \quad |Z_{5\%}| = 1.96$$

$$|Z_{1\%}| > |Z|, \quad |Z_{5\%}| > |Z|$$

∴ accepts H_0 .

10. In a city a sample of thousand people was taken and out of them 540 are vegetarians and the rest are non-vegetarians. Can we say that both habits of eating are equally eating popular in the city or not.

i). 1% level of significance.

ii). 5% level of significance.

Sol:

$$n = 1000.$$

$$p = \frac{540}{1000} = 0.54.$$

$$P = \frac{1}{2}, Q = \frac{1}{2}.$$

H_0 : both habits are equally popular in city.

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$$

$$= \frac{0.54 - 0.5}{\sqrt{\frac{0.5 \cdot 0.5}{1000}}}$$

$$= \frac{0.04}{0.01581}$$

$$Z = 2.5298$$

i). 1% level of significance.

$$|Z_{1\%}| = 2.58.$$

$$|Z_{1\%}| > |Z|$$

Hence accepts H_0 .

ii). 5% level of significance.

$$|Z_{5\%}| = 1.96.$$

$$|Z_{5\%}| < |Z|.$$

Hence rejects H_0 .

11. A salesman in a departmental stores claims that atmost 60% of the shoppers enter the store, leaves without making a purchase. A Random sample of 50 shoppers shown that 35 of them left without making a purchase. Are these sample result consistent with the claim of the salesman? Use the level of significance of 0.05.

atmost 60%, $\leq 60\%$.

Sol:

$$H_0: p \leq P$$

$$H_1: p > P \text{ (Right)}, p = 0.7, P = 0.6.$$

$$p = \frac{35}{50}$$

Level of significance, 5%.

$$|Z_{5\%}| = 1.645$$

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.7 - 0.6}{\sqrt{\frac{0.6 \times 0.4}{50}}} = 1.443$$

$$|Z| < |Z_{5\%}|$$

$\therefore H_0$ is accepted.

Q5) A biased coin was tossed 400 times and 240 heads turned up. (Find the probability of throwing heads in a single trial at most lies b/w 0.53 & 0.67) show that confidence limits

Sol:

$$p = \frac{240}{400} = 0.6$$

$$q = 1 - 0.6 = 0.4$$

$$P = p \pm 3 \sqrt{\frac{pq}{n}}$$

$$= 0.6 \pm 3 \sqrt{\frac{0.6 \times 0.4}{400}}$$

$$= 0.6 \pm 3 \sqrt{6 \times 10^{-4}}$$

$$= 0.6 \pm (0.024) \times 3$$

$$P = 0.6 \pm 0.0734$$

confidence limits ; $0.6 + 0.0734$, $0.6 - 0.0734$

$$: 0.6734, 0.5266$$

① Testing of significance for difference of proportions.

n_1, n_2 - sample size.

p_1, p_2 - sample proportions.

$$Z = \frac{p_1 - p_2}{\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

Q6) In a Rural area where no development was undertaken 160 out of a sample of 250 farmers were debited. In an other area where development work was in progress 84 out of a sample of 150 farmers were indebted. Would you consider that the latter area is enjoying greater property as indebted by a low lower percentage of indebted.

Sol:

H_0 : No significant difference.

$$p_1 = \frac{160}{250} = 0.64$$

$$n_1 = 250$$

$$p_2 = \frac{84}{150} = 0.56$$

$$n_2 = 150$$

$$P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

$$= \frac{(250)(0.64) + (150)(0.56)}{250 + 150}$$

$$= \frac{160 + 84}{400}$$

$$= \frac{244}{400}$$

$$P = 0.61$$

$$Q = 1 - P = 1 - 0.61 = 0.39$$

$$Z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$= \frac{0.64 - 0.56}{\sqrt{(0.61)(0.39)\left(\frac{1}{250} + \frac{1}{150}\right)}}$$

$$= \frac{0.08}{\sqrt{(0.2379)(0.01066)}}$$

$$= \frac{0.08}{\sqrt{0.0025376 \times 10^{-3}}}$$

$$= \frac{0.08}{0.0503}$$

$$Z = 1.5881$$

we choose 5% level of significant.

$$|Z_{5\%}| = 1.96$$

$$|Z| < |Z_{5\%}|$$

Hence accept H_0 .

(or).

$H_0: n_1 < n_2$ (Right tail)

we choose 5% level of significance.

$$|Z_{5\%}| = 1.645$$

$$|Z| < |Z_{5\%}|$$

Hence accept H_0 .

4) In a Random sample of 1000 persons from city coimbatore. 400 are found to be consumers of wheat in a sample of 800 from a city madurai 400 are found to be consumers of wheat. Do this data involve a significant difference between the two cities.

Sol: H_0 : No significant difference

$$p_1 = \frac{400}{1000} = 0.4, n_1 = 1000$$

$$p_2 = \frac{400}{800} = 0.5, n_2 = 800$$

$$P = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2}$$

$$= \frac{(1000)(0.4) + (800)(0.5)}{1000 + 800}$$

$$= \frac{800}{1800}$$

$$P = 0.444$$

$$Q = 1 - 0.444 = 0.555 \approx 0.56$$

$$Z = \frac{p_1 - p_2}{\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{-0.1}{\sqrt{(0.44)(0.56) \left(\frac{1}{1000} + \frac{1}{800} \right)}}$$

$$= \frac{-0.1}{\sqrt{0.2464 (2.25 \times 10^{-3})}}$$

$$= \frac{-0.1}{\sqrt{5.544 \times 10^{-4}}}$$

$$= \frac{-0.1}{0.0235}$$

$$Z = -4.247$$

$$|Z| = 4.247$$

we choose 5% level of significance.

$$|Z_{5\%}| = 1.96 \quad (\text{two tails})$$

$$|Z| > |Z_{5\%}|$$

Hence rejects H_0 .

Random samples of 1000 men, 800 women were asked whether you would like to have a school near the residence. 400 men and 325 women were in favour of the proposal. Test the hypothesis that the proportion of men & women in favour of proposal are same at 5% level of significance.

$$\text{Sol: } H_0: n_1 = n_2$$

$$p_1 = \frac{400}{1000} = 0.4, \quad n_1 = 1000$$

$$p_2 = \frac{325}{800} = 0.406, \quad n_2 = 800$$

$$P = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2}$$

$$= \frac{(400)(0.4) + (800)(0.406)}{1000 + 800}$$

$$= \frac{525}{1800}$$

$$= \frac{525}{1800}$$

$$P = 0.292$$

$$Q = 1 - 0.292 = 0.708$$

$$Z = \frac{p_1 - p_2}{\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$= \frac{0.5 - 0.541}{\sqrt{(0.525)(0.475) \left(\frac{1}{3100} + \frac{1}{600} \right)}}$$

$$= \frac{-0.041}{\sqrt{(0.2493)(4.166 \times 10^{-3})}}$$

$$= \frac{-0.041}{\sqrt{1.039 \times 10^{-3}}}$$

$$= \frac{-0.041}{0.0322}$$

$$Z = -1.271$$

$$|Z| = 1.271$$

we choose 5% level of significant.

$$|Z_{5\%}| = 1.96$$

$$|Z| < |Z_{5\%}|$$

Hence accepts H_0 .

Test of significance for small samples t-Test.

def:

Let $x_1, x_2, \dots, x_n$ be a random sample size n from a normal population with mean μ , variance (σ^2) the

students t-Test is defined in the statistics as $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$

where, $\bar{x} \rightarrow \text{mean} = \frac{\sum x}{n}$

$$s^2 = \frac{1}{n-1} \sum (x - \bar{x})^2$$

(or)

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$$

[degrees of freedom : $n-1$]

where, $\bar{x} = \frac{\sum x}{n}$

$$s^2 = \frac{\sum x^2}{n} - (\bar{x})^2$$

s^2 = sample variance.

σ^2 = population variance.

μ = population variance-mean.

$\bar{x}$ = sample mean.

type-1

16) Sandal powder is packed into packets by a machine a random sample of 12 packets is drawn and their weights are found to be (in kg) 0.49, 0.48, 0.47, 0.48, 0.49, 0.50, 0.51, 0.49, 0.48, 0.50, 0.51, 0.48. Test if the average packing can be taken as 0.5 kg

Sol:

$$n=12$$

$$H_0: \mu = 0.5$$

$$\bar{x} = \frac{\sum x}{n} = \frac{5.88}{12} = 0.49$$

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n-1}}$$

$$s^2 = \frac{\sum x^2}{n} - (\bar{x})^2$$

$$= \frac{2.883}{12} - (0.49)^2$$

$$= 0.24025 - (0.2401)$$

$$s^2 = 1.5 \times 10^{-4} \text{ (or)} s^2 = (1.9 \times 10^{-4})$$

$$s = 0.0132 \text{ (or)} s = 0.01$$

$$t = \frac{0.49 - 0.5}{0.0132 / \sqrt{12-1}}$$

$$t = \frac{-0.01 \times 3.3166}{0.01}$$

$$t = -3.3166 \Rightarrow |t| = 3.3166$$

we choose 5% level of significant with degrees of freedom $(n-1) = 11$.

$$t_{5\%} = 2.20$$

$$|t_{5\%}| < |t| \therefore \text{Hence reject } H_0.$$

0.49	0.2401
0.48	0.2304
0.47	0.2209
0.48	0.2304
0.49	0.2401
0.50	0.25
0.51	0.2601
0.49	0.2401
0.48	0.2304
0.50	0.25
0.51	0.2601
0.48	0.2304
$\sum x = 5.88$	$\sum x^2 = 2.883$

17)

The average breaking strength of steel rods is specified to be 17.5 to test this sample of 14 rods are tested and the following results 15, 18, 16, 21, 19, 21, 17, 17, 15, 17, 20, 19, 17, 18 is the result of the experiment significant? also obtain the 95% confidence interval from the average breaking strength.

$$H_0: \text{No significant difference}$$

Given,

$$\mu = 17.5$$

$$n = 14$$

$$\bar{x} = \frac{\sum x}{n} = \frac{250}{14} = 17.8571$$

$$s^2 = \frac{\sum x^2}{n} - (\bar{x})^2$$

$$= \frac{4514}{14} - (17.8571)^2$$

$$= 322.4285 - 318.8760$$

$$s^2 = 3.5524$$

$$s = 1.8848$$

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n-1}} = \frac{17.8571 - 17.5}{1.8848 / \sqrt{13}}$$

$$t = \frac{0.3571 \times 3.6055}{1.8848}$$

$$t = \frac{1.2875}{1.8848} = 0.6831$$

x	x ²
15	225
18	324
16	256
21	441
19	361
21	441
17	289
17	289
15	225
17	289
20	400
19	361
17	289
18	324
$\sum x = 250$	$\sum x^2 = 4514$

95% confidence co. = 5% of level of significant.

we choose 5% LOS with d.f (n-1) = 13.

$$t_{5\%} = 2.16$$

$$|t| < |t_{5\%}|$$

∴ Hence accept H_0 .

Interval

$$\bar{x} \pm t_{5\%} \left(\frac{s}{\sqrt{n-1}} \right)$$

$$17.8571 \pm (2.16)(0.5226)$$

$$17.8571 \pm 1.1288$$

$$17.8571 + 1.1288, 17.8571 - 1.1288$$

$$\text{Interval: } (18.9859, 16.7283)$$

$$\Downarrow (16.7283, 18.9859)$$

18) A certain medicine administrator to each of 10 patients resulted in the following: increases in the blood pressure 8, 8, 7, 5, 4, 1, 0, 0, -1, -1. can it be concluded that the medicine was responsible for the decrease in that pressure.

Sol:

H_0 : No significant difference.

$$n = 10$$

$$\mu = 0$$

$$\bar{x} = \frac{\sum x}{n} = \frac{31}{10} = 3.1$$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$$

$$s^2 = \frac{\sum x^2}{n} - (\bar{x})^2$$

$$= \frac{121}{10} - (3.1)^2$$

$$= 22.1 - 9.61$$

$$s^2 = 12.49$$

$$s = 3.5341$$

$$t = \frac{3.1 - 0}{3.5341/\sqrt{9}}$$

$$t = \frac{3.1 \times 3}{3.5341} = \frac{9.3}{3.5341}$$

$$|t| = 2.6315$$

we choose 5% of level of significant with d.f (n-1) = 9.

$$|t_{5\%}| = 2.26$$

$$|t_{5\%}| < |t|$$

∴ Hence reject H_0 .

x	x ²
8	64
8	64
7	49
5	25
4	16
1	1
0	0
0	0
-1	1
-1	1
31	321

1. Type - 2

T-test for difference of means of two small samples.

s_1, s_2 are standard deviation of two samples & $\bar{x}_1, \bar{x}_2$ are the means of the two samples

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

9) Two independent samples from normal population with equal variance of the following.

Sample	Size	Mean	S.D
1	16	23.4	2.5
2	12	24.9	2.8

Is the difference b/w the means significant?

Sol:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

$$t = \frac{23.4 - 24.9}{\sqrt{\frac{16(2.5)^2 + 12(2.8)^2}{26} \left(\frac{1}{16} + \frac{1}{12} \right)}}$$

$$= \frac{-1.5}{\sqrt{\frac{194.08}{26} (0.1458)}}$$

$$= \frac{-1.5}{\sqrt{7.4646 (0.1458)}}$$

$$= \frac{-1.5}{\sqrt{1.0883}} = \frac{-1.5}{1.0432}$$

$$t = -1.4378$$

$$|t| = 1.4378$$

we choose 5% LOS with $df = n_1 + n_2 - 2 = 26$.

$$|t_{5\%}| = 2.06$$

$$|t| < t_{5\%}$$

Hence accept H_0

20). A group of 10 rats fed on diet A and another group of 8 rats fed on diet B recorded the following increases in weight (gms)

Diet A 5, 6, 8, 1, 12, 4, 3, 9, 6, 10.

Diet B 2, 3, 6, 8, 10, 1, 2, 8

Does it show superiority of diet A over diet B.

Sol:

$H_0: \mu_1 > \mu_2$ / No significant difference. (left tailed).

$H_1: \mu_1 < \mu_2$.

Given, $n_1 = 10$ $\bar{x}_1 = 64/10 = 6.4$

$n_2 = 8$ $\bar{x}_2 = 40/8 = 5$

$$d.f = n_1 + n_2 - 2$$

$$= 10 + 8 - 2$$

$$= 16.$$

$$s_1^2 = \frac{\sum x_1^2}{n_1} - (\bar{x}_1)^2$$

$$= 51.2 - (6.4)^2$$

$$s_1^2 = 10.24$$

$$s_1 = 3.2$$

$$s_2^2 = \frac{\sum x_2^2}{n_2} - (\bar{x}_2)^2$$

$$= 36.25 - (5)^2$$

$$s_2^2 = 10.25$$

x_1^2	x_2^2
25	4
36	9
64	36
1	64
144	100
16	1
9	14
81	64
36	282
100	
512	

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$t = \frac{6.4 - 5}{\sqrt{\frac{10(10.24) + 8(10.25)}{16} \left(\frac{1}{10} + \frac{1}{8} \right)}}$$

$$= \frac{1.4}{\sqrt{\frac{184.4}{16} (0.225)}}$$

$$= \frac{1.4}{\sqrt{11.525 \times (0.225)}}$$

$$= \frac{1.4}{\sqrt{2.5931}} = \frac{1.4}{1.6103}$$

$$|t| = 0.8693$$

we choose $\alpha = 10\%$ w.r.t d.f = 16.

$$t_{\alpha} = 1.746$$

$$|t| < t_{\alpha}$$

Hence accepts H_0 .

Q). The height of six randomly chosen sailors are [in inches] : 72, 71, 69, 68, 65, 63. Those of ten randomly chosen army men are 73, 72, 71, 70, 69, 69, 66, 65, 62, 61. Analyze the concept that the army men are shorter, on the average than the sailors.

Sol:

$H_0: \mu_2 < \mu_1$ (left tail).

$H_1: \mu_2 > \mu_1$

$n_1 = 6$

$n_2 = 10$

$\bar{x}_1 = 68$

$\bar{x}_2 = 67.8$

$$s_1^2 = \frac{\sum (x_1 - \bar{x}_1)^2}{n_1}$$

$$= \frac{60}{6} = 10$$

$$s_2^2 = \frac{\sum (x_2 - \bar{x}_2)^2}{n_2}$$

$$= \frac{153.64}{10}$$

$$= 15.364$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$= \frac{68 - 67.8}{\sqrt{\frac{6(10) + 10(15.364)}{14} \left(\frac{1}{6} + \frac{1}{10} \right)}}$$

$$= 0.2$$

$$\sqrt{\frac{213.64}{14} (0.2666)}$$

x_1	x_2	$x_1 - x_2$	$x_2 - \bar{x}_2$
72	73	4	5.2
71	72	3	4.2
69	71	1	3.2
68	70	0	2.2
65	69	-3	1.2
63	69	-5	1.2
66	66	0	-1.8
65	65	-2.8	-2.8
62	62	-5.8	-5.8
61	61	-6.8	-6.8
67.8	67.8	0	0

$$(x_1 - \bar{x}_1)^2$$

$$(x_2 - \bar{x}_2)^2$$

$$16$$

$$27.024$$

$$9$$

$$17.64$$

$$1$$

$$10.24$$

$$0$$

$$4.84$$

$$9$$

$$1.44$$

$$25$$

$$1.44$$

$$60$$

$$3.24$$

$$1$$

$$11.84$$

$$1$$

$$33.64$$

$$1$$

$$48.24$$

$$1$$

$$153.64$$

$$= \frac{0.2}{\sqrt{15.36 (0.2666)}}$$

$$= \frac{0.2}{\sqrt{4.0683}} = \frac{0.2}{2.0170}$$

$$= 0.0991$$

$$|t| = 0.0991$$

we choose 5% i.e. 0.05 with d.f. = $n_1 + n_2 - 2$ = 14.

$$t_{5\%} = 1.761$$

$$|t| < t_{5\%}$$

Hence accepts H_0 .

properties of t-distribution:

1. The variable ranges from $-\infty$ to $+\infty$.

2. t distribution is symmetrical about $t=0$ and has a mean zero.

3. Variance = $\frac{v}{v-2}$ if $v > 2$, $\mu_2 > 1$ always.

F-test of significance for small sample.

If s_1^2 and s_2^2 are the variance of two sample of size n_1 & n_2 respectively.

The estimate of the population

variance based on these samples are

respectively $s_1^2 = \frac{n_1 s_1^2}{n_1 - 1}$ and $s_2^2 = \frac{n_2 s_2^2}{n_2 - 1}$

The quantities $V_1 = n_1 - 1$ and $V_2 = n_2 - 1$ are called the degrees of freedom of these estimators.

In setting $F = \frac{s_1^2}{s_2^2}$, the numerator is greater than denominator, so $F > 1$.

$$F = \frac{s_1^2}{s_2^2} = \frac{n_1 s_1^2 (n_2 - 1)}{(n_1 - 1) n_2 s_2^2}$$

where s_1^2, s_2^2 of two samples and

$$s_1^2 = \frac{\sum (x - \bar{x})^2}{n_1 - 1}$$

$$s_2^2 = \frac{\sum (x - \bar{x})^2}{n_2 - 1}$$

if $s_1^2 > s_2^2$, if $s_2^2 > s_1^2$

$$F = \frac{s_1^2}{s_2^2}$$

$$F = \frac{s_2^2}{s_1^2}$$

22). In a sample of 8 observations the sum of the squares of deviation of the sample values from the sample mean was 94.5 and in another sample 10 observations, it was found to be 101.7. Test whether the difference of variance is significant.

H₀: No significant difference.

Sol:

$$n_1 = 8$$

$$\sum (x_1 - \bar{x}_1)^2 = 94.5$$

$$s_1^2 = \frac{\sum (x_1 - \bar{x}_1)^2}{n_1 - 1} = \frac{94.5}{8 - 1} = 13.5$$

$$s_2^2 = \frac{\sum (x_2 - \bar{x}_2)^2}{n_2 - 1} = \frac{101.7}{10 - 1} = 11.3$$

$$F = \frac{s_1^2}{s_2^2}, s_1^2 > s_2^2$$

$$= \frac{13.5}{11.3}$$

$$F = 1.1946$$

$$d.f = (n_1 - 1, n_2 - 2) = (8 - 1, 10 - 1)$$

$$d.f = (7, 9)$$

we choose 5% LOS.

$$F_{5\%}(7, 9) = 3.29$$

$$|F| < F_{5\%}$$

Hence accept H₀.

23). In one sample of 10 observations, sum of the squares of deviations of the sample values from sample mean was 120 & another sample of 12 observations it was 244.

Test whether this difference is significant at 5% LOS.

sol: H_0 : No significant difference.

$$n_1 = 10 \quad n_2 = 10$$

$$\sum (x_1 - \bar{x}_1)^2 = 120 \quad \sum (x_2 - \bar{x}_2)^2 = 314$$

$$s_1^2 = \frac{\sum (x_1 - \bar{x}_1)^2}{n_1 - 1} = \frac{120}{10 - 1} = 13.333$$

$$s_2^2 = \frac{\sum (x_2 - \bar{x}_2)^2}{n_2 - 1} = \frac{314}{10 - 1} = 34.885$$

$$F = \frac{s_2^2}{s_1^2} = \frac{34.885}{13.33} \quad s_2^2 > s_1^2$$

$$F = 2.61409$$

$$d.o.f = (n_1 - 1, n_2 - 1) = (9, 9)$$

$$F_{5\%}(9, 9) = 2.90$$

$$|F| < F_{5\%}$$

$\therefore$ Hence accept H_0 .

24. A group of 10 rats fed on diet A and another group of 8 rats fed on different diet B, recorded the following increase in weight.

Diet A 5, 6, 8, 1, 10, 4, 3, 9, 6, 10 gms.

Diet B 2, 3, 6, 8, 1, 10, 2, 8, gms.

Find the variance significantly different. H_0 : No significant difference.

sol: $n_1 = 10$
 $n_2 = 8$

$$\bar{x}_1 = \frac{\sum x_1}{n_1} = \frac{64}{10} = 6.4$$

$$\bar{x}_2 = \frac{\sum x_2}{n_2} = \frac{40}{8} = 5$$

refer (20)

$$d.f = (n_1 - 1, n_2 - 1)$$

$$= (9, 7)$$

$$s_1^2 = \frac{\sum x_1^2}{n_1} - (\bar{x}_1)^2$$

$$= \frac{512}{10} - (6.4)^2$$

$$s_1^2 = 10.24$$

$$s_2^2 = \frac{\sum x_2^2}{n_2} - (\bar{x}_2)^2$$

$$= \frac{250}{8} - (5)^2$$

$$s_2^2 = 10.25$$

$$F = \frac{n_1 s_1^2 (n_2 - 1)}{(n_1 - 1) n_2 s_2^2}$$

$$= \frac{10 (10.24) 7}{(9) 8 (10.25)}$$

$$= \frac{716.8}{728} = \frac{1}{0.9712} = 1.029$$

$$s_1^2 = \frac{10(10.24)}{10-1} = 11.377$$

$$s_2^2 = \frac{n_2 s_2^2}{n_2 - 1} = \frac{8(10.25)}{8-1} = 11.714$$

$$F = \frac{s_2^2}{s_1^2}$$

$$= \frac{11.714}{11.377}$$

$$F = 1.0295$$

we choose 5% LOS.

$$F_{5\%}(9,7) = 3.68$$

$$|F| < F_{5\%}$$

$\therefore$ Hence accept H_0 .

25. From the following two samples values find out whether they have come from the same population.

Sample 1: 17, 27, 18, 25, 21, 29, 27, 23, 17.

Sample 2: 16, 16, 20, 16, 20, 17, 15, 21.

(difference of two means of two sample

same population

population with equal variances)

Key words for F-test.

CHI - square test:

1. chi-square test of goodness of fit.

If O_i ($i=1, 2, \dots, n$) is a set of observed (experimental) frequencies and

E_i ($i=1, 2, \dots, n$) is the corresponding set of expected (theoretical or hypothetical) frequencies, then

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

condition $\sum_{i=1}^n O_i = \sum_{i=1}^n E_i$ follows chi-square

distribution with $(n-1)$ degrees of freedom.

Independent of attributes:

To test independence of attributes

we calculate χ^2 (χ^2) value from the cell frequencies and compare with table

value of χ^2 corresponding to degree of

freedom.

For a $a \times b$ contingency table $\frac{a/b}{c/d}$,

the χ^2 value is

$$\chi^2 = \frac{(ad-bc)^2}{(a+c)(b+d)(a+b)(c+d)}$$

Degree of freedom is $(a-1)(b-1) = 1$.

In the case of 6×9 contingency table
degrees of freedom is $(6-1) \times (9-1)$

26) A die was thrown 148 times. Denoting
 x to be the number appearing on the
top face to it, the observed frequency
of x is given below.

x	1	2	3	4	5	6
f	69	18	85	82	86	98

what opinion you would form for
the accuracy of the die?

Sol
 H_0 : die is unbiased.

$$\chi^2 = \frac{\sum (O_i - E_i)^2}{E_i}$$

$$E_i = \frac{1498}{6} = 83.$$

$$\chi^2 = \frac{(69-83)^2 + (18-83)^2 + (85-83)^2 + (82-83)^2 + (86-83)^2 + (98-83)^2}{83}$$

$$= \frac{(96 + 25 + 4 + 1 + 9 + 225)}{83}$$

$$\chi^2 = \frac{460}{83} = 5.5421$$

we choose 5% LOS.
degrees of freedom = $n-1 = 6-1 = 5$.

$$\chi^2_{5\%} = 11.070.$$

$$\chi^2 < \chi^2_{5\%}$$

$\therefore$ Hence accept H_0 .

The table below gives the no. of
aircraft accidents that occurred during
the various days of the week. Test
whether the accidents are uniformly
distributed over the week.

Days	Mon	Tues	Wed	Thurs	Fri	Sat
no. of accidents.	14	18	12	11	15	14

Sol
 H_0 : No. of accidents distributed over a week.

$$E_i = \frac{84}{6} = 14.$$

$$\chi^2 = \frac{(14-14)^2 + (18-14)^2 + (12-14)^2 + (11-14)^2 + (15-14)^2 + (14-14)^2}{14}$$

$$= \frac{0 + 16 + 4 + 9 + 1 + 0}{14}$$

$$\chi^2 = \frac{30}{14}$$

$$\chi^2 = 2.1428.$$

we choose 5% LOS with d.f = 6-1 = 5

$$\chi^2_{5\%} = 11.070$$

$$\chi^2 < \chi^2_{5\%}$$

∴ Hence accept H_0 .

28) Find (it there is any association b/w extravagance in father and extravagance in sons) from the following data.

	Extravagant father	Miserly father
Extravagant son	837	741
Miserly son	545	234

Determine the coefficient of association

also.

Sol:

H_0 :

$$\chi^2 = \frac{(ad-bc)^2 (a+b+c+d)}{(a+c)(b+d)(a+b)(c+d)}$$

$$= \frac{[837 \times 234 - (545 \times 741)]^2}{(837 + 234 + 545 + 741)(837 + 545)(741 + 234)(545 + 234)}$$

$$= \frac{1.0714 \times 10^{11} (1847)}{7.0734 \times 10^{11}}$$

$$= 0.15146 \times 1847$$

$$\chi^2 = 279.746$$

$$d.f = (r-1)(c-1) = (2-1)(2-1) = 1$$

we choose 5% LOS with d.o.f = 1

$$\chi^2_{5\%} = 3.841$$

$$\chi^2 > \chi^2_{5\%}$$

Hence rejects H_0 .

29) on the basis of information noted below find out whether the new treatment is comparatively superior to conventional one.

	Favourable	Non-favourable	total
conventional	40 a	70 b	110
New	60 c	20 d	80
total	100	90	190

Sol:

H_0 : New treatment is comparatively superior.

$$\psi^2 = \frac{(ad-bc)^2 (a+b+c+d)}{(a+c)(b+d)(a+b)(c+d)}$$

$$= \frac{(110 \times 30 - 60 \times 70)^2 (200)}{(100)(100)(110)(90)}$$

$$= \frac{9 \times 10^6 \times 200}{99 \times 10^6}$$

$$\psi^2 = 18.1818$$

we choose 10% 5% with d.f = 1.

$$\psi^2_{5\%} = 3.841$$

$$\psi^2_{5\%} < \psi^2$$

Hence rejects H_0

30) In an investigation into the health & nutrition of two groups of children of different social status the following results are got.

Social status \ Health	Poor	Rich	Total
Below normal	130	20	150
Normal	103	108	210
Above Normal	54	96	150
Total	287	224	511

Discuss the relation b/w the health & their social status.

Sol H_0 : health and nutrition of two groups of children of different social status.

Expected frequency (E_i).

Health \ SS	Poor	Rich	Total
below normal	$\frac{256 \times 150}{480} = 80$	$\frac{224 \times 150}{480} = 70$	150
normal	$\frac{256 \times 210}{480} = 112$	$\frac{224 \times 210}{480} = 98$	210
above normal	$\frac{256 \times 120}{480} = 64$	$\frac{224 \times 120}{480} = 56$	120
total	256	224	480

$$\psi^2 = \frac{\sum (O_i - E_i)^2}{E_i}$$

$$= \frac{(130-80)^2}{80} + \frac{(20-70)^2}{70} + \frac{(103-112)^2}{112} + \frac{(108-98)^2}{98}$$

$$+ \frac{(54-64)^2}{64} + \frac{(96-56)^2}{56}$$

$$= \frac{2500}{80} + \frac{2500}{70} + \frac{100}{112} + \frac{100}{98} + \frac{1600}{64} + \frac{1600}{56}$$

$$= 21.25 + 35.7142 + 0.8958 + 1.0204 + 1.0204$$

$$= 118.8774$$

$$\chi^2 = 122.4488$$

$$d.f = (r-1)(c-1) = (3-1)(2-1) = 2$$

we choose $\alpha = 5\%$ for with $d.f = 2$.

$$\chi^2_{\alpha, d.f} = 5.991$$

$$\chi^2 < \chi^2_{\alpha, d.f}$$

$\therefore$ Hence rejects H_0